

EAI — Study Timeline

6:00 AM → 10:45 AM (Sep 12) | Exam window: 10:45 AM–12:30 PM | Includes 45 min to get ready + 15 min commute before the exam | Do's and Don'ts backed by cited learning-science and sleep research

4h45m total, cleanly split: ~3h45m of actual study/review, then a fixed 1-hour block (45 min getting ready + 15 min commute) ending exactly at exam start. All 10 confirmed sub-questions (Q1–Q5, both sides each) are covered with real active-recall practice before the final self-test.

Hour-by-Hour Timeline

Time	Activity	Why (research basis)
6:00–6:10 AM	Plan check: skim the Study Priority PDF structure (Q1–Q5, both sides each).	Brief upfront planning improves self-regulated study efficiency.
6:10–7:10 AM	Study Block 1 (60 min) — Q1 & Q2 (Intelligent IoT vs Cloud IoT + applications, ML/DL tools/frameworks + deployment/inferencing platforms). Write each from memory, then check.	Active recall (retrieval practice) produces far better long-term retention than re-reading notes — Roediger & Karpicke (2006), Psychological Science.
7:10–7:20 AM	Break — stand, stretch, no screens.	Short breaks help sustain attention.
7:20–8:20 AM	Study Block 2 (60 min) — Q3 & Q4 (TinyML hardware boards + model compression, Jetson Nano + MC/AI-accelerator hardware). Mix the board-name lists with the conceptual compression techniques.	Interleaving different content types improves later recall under exam conditions — interleaving-effect research (Rohrer & Taylor).
8:20–8:30 AM	Break.	Prevents fatigue-driven drop in encoding quality.
8:30–9:15 AM	Study Block 3 (45 min) — Q5 (EML basic blocks + examples, data engineering steps: cleaning/organizing/transformation). Then a full re-skim of all 5 questions' OR sides.	Completes all 10 confirmed sub-questions with retrieval practice before the final review.
9:15–9:45 AM	Self-test — write out all 10 answers from memory (both sides of all 5 questions), full sentences, as if sitting the actual exam.	A dedicated retrieval session under exam-like conditions is what drives the testing effect's retention benefit — Roediger & Karpicke (2006).
9:45–10:30 AM	Get ready — wash up, dress, breakfast/hydration, pack bag (ID, stationery, admit card).	Stabilizes blood glucose for concentration; fixed prep time avoids rushing.
10:30–10:45 AM	Commute to the exam hall. Once there, breathe and settle — avoid anxious last-minute topic discussion with classmates.	Comparing knowledge gaps with peers right before an exam tends to raise anxiety without adding useful recall.

(Table continues to next page)

Do's — Backed by Research

1. Use active recall (self-testing) instead of re-reading notes.

Roediger & Karpicke (2006), Psychological Science / Perspectives on Psychological Science: students who practiced free recall retained significantly more on delayed tests than students who simply re-studied the same material, even though re-studiers felt more prepared.

2. Space topics across separate blocks rather than massing them into one continuous stretch.

Cepeda, Pashler, Vul, Wixted & Rohrer (2006), Psychological Bulletin — a meta-analysis of 839 assessments across 317 experiments confirmed spaced (distributed) practice beats massed practice for retention, regardless of how long until the test.

3. Interleave different topic types instead of drilling one topic repeatedly before moving on.

Interleaving research (e.g. Rohrer & Taylor) shows mixing related-but-different problem types improves the ability to later distinguish and correctly apply the right method under exam conditions.

4. Take short breaks every 60–90 minutes of focused work.

Supports sustained attention and reduces the fatigue-driven decline in encoding quality seen in long unbroken study sessions.

5. Protect a full night's sleep before the exam — don't trade it for one more study block.

Sleep is when declarative memories (facts, procedures, definitions) are consolidated. Reviews of student sleep vs. exam performance consistently find all-nighter cramblers score lower than peers who slept, even when the cramblers logged more total study hours.

6. If you're genuinely short on sleep, a 20–26 minute nap can help — no longer.

Rosekind et al. (1995), NASA Ames Research Center: a controlled 26-minute nap in fatigued pilots improved task performance by 34% and physiological alertness by 54% versus no rest. Naps beyond ~30 min risk entering deep sleep and causing grogginess (sleep inertia) on waking.

7. Stop caffeine at least 6 hours before you intend to sleep.

Drake, Roehrs, Shambroom & Roth (2013), Journal of Clinical Sleep Medicine: caffeine taken even 6 hours before bedtime significantly reduced total sleep time versus placebo.

8. Dim lights and stop using phone/laptop screens 30–60 minutes before sleep.

Chang, Aeschbach, Duffy & Czeisler (2015), PNAS: reading on a light-emitting device before bed suppressed melatonin, delayed sleep onset, reduced REM sleep, and lowered next-morning alertness compared with reading a printed book.

9. Do a short retrieval self-test right after waking, on just your top-ranked topics.

Consolidation continues overnight; a brief retrieval session soon after waking reinforces that consolidated material using the same testing-effect mechanism as evening study.

Don'ts — Backed by Research

1. Don't pull an all-nighter.

Multiple reviews of student sleep research find all-nighters impair both encoding of new material and accurate self-assessment — sleep-deprived students often feel confident despite performing worse.

2. Don't just re-read or highlight your notes as your main study method.

Karpicke, Butler & Roediger (2009) found students overwhelmingly prefer re-reading even though it produces a well-documented “illusion of competence” — it feels like learning without producing comparable retention gains.

3. Don't cram everything into one long unbroken session the night before.

Massed practice consistently underperforms spaced/distributed practice in the Cepeda et al. (2006) meta-analysis, regardless of subject matter.

4. Don't use caffeine or energy drinks late at night to “push through.”

Drake et al. (2013) found caffeine reduces total sleep time even when taken hours before bed, and sleep-deprived people are poor judges of their own impairment.

5. Don't scroll on your phone right before trying to sleep.

Chang et al. (2015) and related studies (Chinoy et al., 2018) show light-emitting screens delay melatonin release and shift the circadian clock, making it harder to fall asleep even if you feel tired.

6. Don't skip meals or stay dehydrated during the study session.

Blood glucose and hydration affect sustained attention and working memory during long study sessions.

7. Don't engage in anxious last-minute topic discussions with classmates outside the exam hall.

This kind of social comparison right before a test tends to raise anxiety without adding useful retrieval practice at that point.

Key sources cited above:

- Roediger, H. L., & Karpicke, J. D. (2006). The Power of Testing Memory: Basic Research and Implications for Educational Practice. *Perspectives on Psychological Science*, 1(3), 181–210.
- Karpicke, J. D., Butler, A. C., & Roediger, H. L. (2009). Metacognitive strategies in student learning: Do students practise retrieval when they study on their own? *Memory*, 17(4), 471–479.
- Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin*, 132(3), 354–380.
- Rosekind, M. R., et al. (1995). Crew Factors in Flight Operations 9: Effects of Planned Cockpit Rest on Crew Performance and Alertness in Long-Haul Operations. NASA Ames Research Center Technical Memorandum; summarized in Rosekind et al., *Journal of Sleep Research*.
- Drake, C., Roehrs, T., Shambroom, J., & Roth, T. (2013). Caffeine Effects on Sleep Taken 0, 3, or 6 Hours before Going to Bed. *Journal of Clinical Sleep Medicine*, 9(11), 1195–1200.
- Chang, A. M., Aeschbach, D., Duffy, J. F., & Czeisler, C. A. (2015). Evening use of light-emitting eReaders negatively affects sleep, circadian timing, and next-morning alertness. *PNAS*, 112(4), 1232–1237.